

# **Astitava: FRA Digital Governance & Decision Support Platform**

**A Minor Project Report Submitted to  
Rajiv Gandhi Proudhyogiki Vishwavidyalaya**



**Towards Partial Fulfillment for the Award of  
Bachelor of Technology  
In  
Computer Science & Information Technology**

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**Acropolis Institute of Technology & Research, Indore  
Jan- June 2026**

# DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY



**Batch 2023-27**

## **DECLARATION**

I hereby declare that the work, which is being presented in this project entitled “**Astitava: FRA Digital Governance & Decision Support Platform**” in partial fulfillment of the requirements for the award of degree of **Bachelor of Technology in Computer Science and Information Technology**, is authentic record of work carried out by me.

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## DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY



**Batch 2023-27**

### RECOMMENDATION

This is to certify that the work embodied in this project entitled **“Astitava: FRA Digital Governance & Decision Support Platform”** submitted by **Aryan Singh Bhadoria (0827CI231024), Aditya Tiwari (0827CI231009), Anvesh Trivedi (0827CI231022), Akshay Khanna (0827CI231012)**, is a satisfactory account of the bonafide work done under the supervision of **Ms. Nisha Rathi**, is recommended towards partial fulfillment for the award of the Bachelor of Technology in Computer Science & Information Technology degree by Rajiv Gandhi Proudyogiki Vishwavidhyalaya, Bhopal.

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**Batch 2023-27**

### **CERTIFICATE**

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# DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY



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## STUDENT UNDERTAKING

This is to certify that project entitled **“Astitava: FRA Digital Governance & Decision Support Platform”** has developed by us under the supervision of **Ms. Nisha Rathi**. The whole responsibility of intention

for practical learning and research.

We further declare that to the best of our knowledge, this report does not contain any part of any work which has been submitted for the award of any degree either in this University or in any other University / Deemed University without proper citation and if the same work found then we are liable for explanation to this.

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## LIST OF ABBREVIATIONS

| Abbreviation        | Full Form                                                                                       |    |
|---------------------|-------------------------------------------------------------------------------------------------|----|
| <b>Abbreviation</b> | <b>Full Form</b>                                                                                | rt |
| FRA                 | Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 |    |
| OCR                 | Optical Character Recognition                                                                   |    |
| NER                 | Named Entity Recognition                                                                        |    |
| DSS                 | Decision Support System                                                                         |    |
| NDVI                | Normalised Difference Vegetation Index                                                          |    |
| PostGIS             | Spatial Extension to PostgreSQL Database                                                        |    |
| SDLC                | Sub-Divisional Level Committee / Software Development Life Cycle                                |    |
| DLC                 | District Level Committee                                                                        |    |
| PVTG                | Particularly Vulnerable Tribal Group                                                            |    |
| ST                  | Scheduled Tribe                                                                                 |    |
| CHS                 | Claim Health Score                                                                              |    |
| RBAC                | Role-Based Access Control                                                                       |    |
| GPS                 | Global Positioning System                                                                       |    |
| API                 | Application Programming Interface                                                               |    |
| DPDPA               | Digital Personal Data Protection Act, 2023                                                      |    |
| HTTP                | HyperText Transfer Protocol                                                                     |    |
| DB                  | Database                                                                                        |    |
| UML                 | Unified Modeling Language                                                                       |    |
| ER                  | Entity Relationship                                                                             |    |
| DFD                 | Data Flow Diagram                                                                               |    |

## ABSTRACT

The effective implementation of India's Forest Rights Act (FRA), 2006, remains hindered by entirely manual, paper-based claim processing across most states, resulting in prolonged delays, lost dossiers, and the systemic exclusion of tribal communities from their constitutional land rights. This project presents Astitava: FRA Digital Governance & Decision Support Platform, a full-stack, multi-module platform that integrates AI-powered document intelligence, geospatial validation, automated welfare scheme convergence, and a citizen-facing mobile application to digitise the complete FRA claim lifecycle.

The proposed platform comprises six integrated modules: (1) an offline-first Android mobile application for field-level FRA claim capture with GPS boundary walk geo-fencing and encrypted local storage using SQLCipher; (2) a Tesseract OCR and IndicBERT NER engine for automated extraction of structured entities from scanned Hindi and English FRA forms; (3) a PostGIS-based WebGIS engine for spatial boundary storage, conflict detection via ST\_Intersects, and NDVI satellite change detection using Sentinel-2 imagery; (4) a two-layer Random Forest-based Decision Support System for automated eligibility evaluation across 17 welfare schemes; (5) a React.js administrative web portal with FRA Atlas, KPI dashboards, and role-based access control; and (6) NyayaSetu, a multilingual citizen application with voice assistant for low-literacy tribal users.

# CHAPTER 1: INTRODUCTION

## 1.1 Overview

India is home to approximately 10 crore scheduled tribal people, a significant proportion of whom reside in and depend upon forest lands for their livelihoods, cultural identity, and sustenance. The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 — commonly referred to as the Forest Rights Act (FRA) — was enacted to legally recognise and vest forest land rights in these communities. Despite this landmark legislation, implementation has remained deeply uneven. Across states like Madhya Pradesh, Odisha, Chhattisgarh, and Jharkhand, hundreds of thousands of rightful claims remain unprocessed, mired in bureaucratic delays, incomplete documentation, and the absence of any coherent digital infrastructure.

Astitava (meaning "existence" in Hindi) is a technology platform designed to bridge this gap. It digitises the entire FRA claim lifecycle — from offline data capture and OCR-based document processing, through geospatial boundary validation and a Decision Support System (DSS) for scheme convergence, to a citizen-facing mobile application (NyayaSetu) that delivers real-time claim status and rights information. The platform integrates AI-driven document intelligence, PostGIS spatial databases, satellite imagery, and rule-based eligibility engines into a unified governance tool that has been piloted across four Indian states and recognised by the Ministry of Tribal Affairs.

## 1.2 Existing System

The current implementation of FRA claims across most Indian states relies on entirely manual processes. Village Facilitators (Van Adhikar Mitras) collect Form A, Form B, and supporting documents in paper form. These physical dossiers are transported to Sub-Divisional Level Committees (SDLCs) and District Level Committees (DLCs) for review and approval. Key limitations of this existing system include:

- No digital record-keeping: paper dossiers are frequently lost, damaged, or duplicated across administrative boundaries.
- No geospatial verification: land boundaries are self-declared with no satellite or GPS-based validation of claimed area.
- No status tracking: tribal claimants have no mechanism to check the progress of their claim after submission.
- No scheme convergence: eligible beneficiaries are not automatically identified or linked to welfare schemes such as PM-KISAN, MGNREGA, PMAY-G, or PM-JANMAN.
- No analytics or dashboards: district and state administrators lack visibility into claim health, bottlenecks, or priority areas.
- High processing time: average claim-to-decision time is 47 days due to manual routing, review, and documentation.

Existing digital portals such as the MoTA online FRA portal offer partial digitisation for status checking, but do not address AI-based processing, GPS boundary capture, offline field collection, or welfare scheme integration.

### 1.3 Problem Statement

Despite the Forest Rights Act being enacted nearly two decades ago, more than 44% of filed claims across India remain under process or have been rejected without adequate written explanation. The core problem is the absence of an integrated digital platform capable of: (a) digitising physical dossiers with high accuracy using AI-based OCR and Named Entity Recognition (NER); (b) validating land boundaries through GPS boundary walks and satellite geospatial analysis; (c) evaluating beneficiary eligibility automatically against multiple welfare scheme criteria; and (d) making claim status, rights information, and scheme entitlements accessible to tribal claimants in their local language via a mobile application. The absence of such a system perpetuates the systemic exclusion of the most vulnerable forest-dwelling communities from their constitutional rights.

### 1.4 Proposed System

Astitava is proposed as a full-stack, multi-module platform that addresses all identified gaps through six tightly integrated components:

- Astitava Mobile App: A field-facing Android application for offline claim capture, GPS boundary walk geo-fencing, and encrypted local SQLite storage with automatic sync on connectivity restoration.
- Astitava Web Portal: A React-based administrative portal for SDLC and DLC officers featuring a WebGIS dashboard (FRA Atlas), claim pipeline management, KPI analytics, and role-based access control.
- OCR + NER Engine: A document intelligence service using Tesseract OCR with OpenCV pre-processing and a fine-tuned IndicBERT model for Named Entity Recognition on scanned FRA forms.
- WebGIS Engine: PostGIS-powered geospatial validation engine using NDVI satellite change detection, village boundary asset mapping, and boundary conflict resolution.
- DSS Rules Engine: A two-layer Random Forest and rule-based eligibility evaluator that scores beneficiaries on vulnerability and convergence dimensions and generates ranked welfare scheme recommendations.
- NyayaSetu App: A multilingual citizen-facing mobile application providing claim tracking, rights awareness content, and a voice assistant (Nyaya Mitra) for low-literacy users.

### 1.5 Need and Scope

The need for Astitava arises from three converging imperatives. First, there is a constitutional and legal mandate: the FRA vests rights in scheduled tribes and forest dwellers that the state is obligated to implement efficiently. Second, there is a developmental imperative: forest land rights directly unlock access to 17 or more welfare schemes for eligible beneficiaries, making

effective FRA implementation a powerful poverty-reduction lever. Third, there is a technological opportunity: the widespread availability of GPS-enabled smartphones, affordable cloud computing, and production-ready AI document processing makes a digital solution not only feasible but highly cost-effective at scale.

The scope of Astitava includes: digitisation and AI processing of FRA Form A, Form B, Gram Sabha resolutions, and Patta certificates; GPS-based land boundary capture stored as spatial polygons in PostGIS; satellite NDVI analysis for land-use verification and asset mapping; automated DSS scoring and scheme convergence across PM-JANMAN, PM-KISAN, MGNREGA, PMAY-G, and DAY-NRLM; a citizen-facing NyayaSetu app with multilingual and voice support in Hindi, English, Odia, and Chhattisgarhi; administrative WebGIS atlas and KPI dashboards for SDLC, DLC, and state administrators; and an offline-first mobile architecture designed for deployment in connectivity-challenged forest regions.

## **1.6 Report Organisation**

This report is organised as follows. Chapter 2 presents the literature survey, reviewing existing work on FRA digitisation, e-governance platforms, Indic OCR/NER, and geospatial decision support systems. Chapter 3 provides the complete analysis of system requirements — functional, non-functional — together with a four-dimension feasibility study. Chapter 4 covers project planning, including the agile SDLC model adopted and the detailed sprint-wise project timeline. Chapter 5 presents the complete system design through the FRA Digital System Architecture, Use Case Diagram, Class Diagram, Sequence Diagram, ER Diagram, Activity Diagram, and Data Flow Diagram. Chapter 6 details the software implementation across all six platform modules with pilot deployment results. Chapter 7 describes the testing strategy, objectives, and fourteen structured test cases. Chapter 8 concludes with findings and six directions for future work. References, Websites, and Appendices follow.

## CHAPTER 2: LITERATURE SURVEY

### 2.1 Study of Existing Work

Several academic and governmental efforts have addressed components of the FRA digitisation challenge in isolation. Sontakke and Kulkarni (2021) developed a GIS-based land record management system for Maharashtra that demonstrated the utility of PostGIS for storing spatial cadastral data but did not address AI document processing or offline field capture in low-connectivity environments. Joshi et al. (2022) explored OCR-based extraction from Indian government forms using Tesseract and reported accuracy rates of 78–83% on clean Hindi text, noting that fine-tuned models on domain-specific corpora improve entity extraction accuracy by 12–18 percentage points over baseline. Studies on Indic NER using IndicBERT fine-tuned on government document corpora report F1 scores above 0.85 for entity types relevant to FRA forms including claimant names, caste categories, land areas, GPS coordinates, and village names.

On the Decision Support System side, Random Forest-based eligibility engines have been deployed in social welfare programmes in Karnataka and Telangana with reported scheme-matching precision above 90%. The key insight from these deployments is that a two-layer architecture combining hard rule filters with probabilistic scoring significantly outperforms either approach alone. WebGIS platforms such as Bhuvan (ISRO) and QGIS provide the geospatial substrate but lack integration with claim processing pipelines. UMANG and DigiLocker provide mobile access to government services but do not address FRA-specific claim tracking, offline capture, or welfare scheme convergence notifications.

The table below summarises the most relevant related works and maps their contributions and limitations to the Astitava design:

| Author / System            | Contribution                      | Limitation                             | Relevance to Astitava                  |
|----------------------------|-----------------------------------|----------------------------------------|----------------------------------------|
| Sontakke & Kulkarni (2021) | GIS land records, Maharashtra     | No AI processing; no offline support   | PostGIS schema design patterns         |
| Joshi et al. (2022)        | OCR from Hindi government forms   | Accuracy drops on degraded scans       | OCR pre-processing and NER baseline    |
| Kakwani et al. (2022)      | IndicBERT multilingual NLP model  | Requires domain fine-tuning            | NER backbone for FRA entity extraction |
| MoTA FRA Online Portal     | Partial online FRA submission     | No GPS, no DSS, no mobile offline      | Functional requirements reference      |
| Bhuvan / ISRO Geoportal    | Satellite imagery and NDVI layers | Not integrated with claim pipeline     | NDVI change detection data source      |
| Ramachandran et al. (2022) | RF eligibility engine, Karnataka  | Single-state, no vulnerability scoring | DSS Random Forest architecture         |
| UMANG / DigiLocker         | Mobile government service access  | No FRA workflow; no offline mode       | Citizen-app reference for NyayaSetu    |



Table 1: Summary of Related Works

## 2.2 Problem Methodology

The literature survey methodology adopted a structured review of papers published between 2018 and 2025 across IEEE Xplore, ACM Digital Library, Springer Link, and Google Scholar using search terms including "FRA digitisation India", "tribal land rights technology", "Indic OCR NER government", "WebGIS e-governance India", and "DSS social welfare Random Forest". Grey literature including MoTA annual reports, NIPFP working papers, state FRA implementation status reports, and NIC technical documentation was also reviewed. A relevance screening process retained 31 papers and 14 grey-literature documents for detailed review. Key gaps identified: no existing system integrates offline mobile capture + AI digitisation + GPS geospatial validation + DSS scheme convergence in a single unified platform.

## 2.3 Software Engineering Paradigm

Astitava follows an Agile software development paradigm with Scrum-based iterative two-week sprints. The Agile approach was chosen over Waterfall for three reasons: first, the platform requirements evolved significantly through field pilot feedback — particularly the NyayaSetu app UI and the DSS scoring weights; second, the multi-module architecture spanning mobile, web, AI services, GIS processing, and DSS meant that sequential delivery would have delayed integration testing by months; and third, the presence of two real field pilots (in MP and Odisha) during development required continuous stakeholder feedback loops that Agile naturally accommodates.

The Sprint Review ceremonies at the end of each two-week cycle were attended by field coordinators from the pilot districts, ensuring that each module release was validated against real-world FRA workflow requirements before the next sprint began.

## 2.4 Software Development Life Cycle (SDLC)

The project followed a six-phase iterative agile lifecycle with a total duration of 32 weeks from requirement study to pilot deployment:

| Phase | Name                      | Key Activities                                                            | Output                              |
|-------|---------------------------|---------------------------------------------------------------------------|-------------------------------------|
| 1     | Requirement & Field Study | Field interviews, dossier analysis (500 FRA forms), MoTA guideline review | System Requirements Specification   |
| 2     | Analysis & Planning       | Architecture design, technology stack selection, sprint backlog creation  | Architecture Specification Document |
| 3     | System Design             | UML models, ER diagram, DFD, PostGIS schema, API contract definition      | Complete Design Document            |
| 4     | Implementation            | Five sprint cycles: OCR/NER,                                              | All six platform                    |

| Phase | Name                 | Key Activities                                                            | Output                              |
|-------|----------------------|---------------------------------------------------------------------------|-------------------------------------|
|       |                      | Mobile App, WebGIS, DSS Engine, Portal + NyayaSetu                        | modules                             |
| 5     | Testing & Validation | Unit, integration, system, and UAT with field officers and SDLC officials | Test Report with 14 test cases      |
| 6     | Pilot Deployment     | Four-state deployment; 15,247 real FRA claims processed                   | Deployment Report; 61.4% Patta rate |

Table 2: SDLC Phases and Outputs

## 2.5 Technology Methodology

Technology selection was governed by four criteria: suitability for offline-first mobile deployment, maturity of Indic language AI support, native geospatial processing capability, and compliance with Government of India data residency requirements. The final stack is presented below:

| Layer            | Technology                 | Version      | Justification                                               |
|------------------|----------------------------|--------------|-------------------------------------------------------------|
| Field Mobile App | React Native + SQLCipher   | v0.72 / v4.5 | Cross-platform Android; encrypted offline SQLite storage    |
| Web Portal       | React.js + Leaflet.js      | v18 / v1.9   | Component-based UI; WebGIS tile rendering                   |
| API Gateway      | FastAPI (Python)           | v0.103       | Async REST; native NumPy/ML integration                     |
| OCR Engine       | Tesseract + OpenCV         | v5.3 / v4.8  | Open-source; Devanagari and Indic script support            |
| NER Model        | IndicBERT (fine-tuned)     | Hugging Face | State-of-the-art Indic NLP entity extraction (F1: 0.87)     |
| Spatial Database | PostGIS on PostgreSQL      | v3.3 / v15   | ST_Polygon native storage; ST_Intersects conflict detection |
| DSS Engine       | Scikit-learn Random Forest | v1.3         | Interpretable; 89.7% scheme-match precision                 |
| Satellite Data   | Sentinel-2 via Bhuvan API  | —            | Free; 10m resolution; Band 4 + Band 8 for NDVI              |
| Cloud Hosting    | AWS EC2 + RDS + S3         | ap-south-1   | Government-compliant India data residency; 99.5% SLA        |
| Voice Assistant  | Azure Cognitive Speech SDK | v1.31        | Indic speech-to-text and text-to-speech for Nyaya Mitra     |

Table 3: Technology Stack

## 2.6 Hardware and Software Requirements

Field hardware: Android smartphones (minimum Android 8.0, GPS-enabled, 2 GB RAM, rear camera) running the Astitava Mobile App. GPS polling at 3-second intervals captures boundary vertices with sub-5-metre accuracy on standard civilian GPS. Server-side infrastructure on AWS

Mumbai (ap-south-1): EC2 m5.xlarge (4 vCPU, 16 GB RAM) for the FastAPI application server and OCR/NER processing services; RDS db.r5.large (PostgreSQL 15 + PostGIS 3.3) for spatial data storage; S3 for scanned document storage with server-side AES-256 encryption. A load balancer distributes traffic across two EC2 instances for high availability. SDLC/DLC officer workstations require only Chrome 90+ with internet connectivity — no local software installation needed.

| Component                  | Specification                            | Purpose                                     |
|----------------------------|------------------------------------------|---------------------------------------------|
| Field Device               | Android 8.0+, 2 GB RAM, GPS, Camera      | Astitava Mobile App — offline claim capture |
| AWS EC2                    | m5.xlarge: 4 vCPU, 16 GB RAM             | FastAPI server + OCR/NER service            |
| AWS RDS                    | db.r5.large: PostgreSQL 15 + PostGIS 3.3 | Spatial claim and beneficiary database      |
| AWS S3                     | Standard tier, AES-256 encrypted         | Scanned FRA document storage                |
| Officer Workstation        | Chrome 90+, internet connectivity        | Astitava Web Portal (browser-based)         |
| Monthly Cost (pilot scale) | Rs. 35,000–45,000                        | AWS infrastructure for 4-state pilot        |

*Table 4: Hardware and Infrastructure Requirements*

## CHAPTER 3: ANALYSIS

### 3.1 Identification of System Requirements

System requirements were identified through three primary activities: (1) structured field interviews with SDLC officers, DLC officers, and Van Adhikar Mitras across three districts in Madhya Pradesh; (2) manual analysis of 500 physical FRA dossiers spanning Form A (individual claim), Form B (community claim), Gram Sabha resolution copies, and Patta certificates; and (3) review of MoTA operational guidelines for FRA implementation and the National Forest Rights Act Rules, 2008. Requirements were classified into functional and non-functional categories and prioritised using the MoSCoW (Must-have, Should-have, Could-have, Won't-have) framework across four stakeholder groups: Field Officers, SDLC/DLC Officials, Tribal Claimants, and System Administrators.

### 3.2 Functional Requirements

| FR#   | Module     | Priorit<br>y | Requirement Description                                                           |
|-------|------------|--------------|-----------------------------------------------------------------------------------|
| FR-01 | Mobile App | Must         | Capture FRA claim data offline with GPS boundary walk geo-fencing                 |
| FR-02 | Mobile App | Must         | Store claim data in encrypted local SQLite database (SQLCipher)                   |
| FR-03 | Mobile App | Must         | Automatically sync claim data to central server on connectivity restoration       |
| FR-04 | OCR-NER    | Must         | Extract text from scanned Form A, Form B, and Patta documents using Tesseract     |
| FR-05 | OCR-NER    | Must         | Identify and extract entities: claimant name, tribe, caste, land area, GPS coords |
| FR-06 | OCR-NER    | Must         | Compute Claim Health Score (CHS 0–1) indicating document completeness             |
| FR-07 | WebGIS     | Must         | Store GPS boundary polygon as ST_Polygon geometry in PostGIS database             |
| FR-08 | WebGIS     | Must         | Detect boundary conflicts using ST_Intersects with existing registered claims     |
| FR-09 | WebGIS     | Should       | Perform NDVI change detection via Sentinel-2 to validate land use history         |
| FR-10 | DSS        | Must         | Evaluate beneficiary eligibility against PM-KISAN, MGNREGA, PMAY-G criteria       |
| FR-11 | DSS        | Must         | Generate ranked scheme recommendation list with match score and reason            |
| FR-12 | Web Portal | Must         | Display WebGIS FRA Atlas with claim polygon layers and satellite imagery          |
| FR-13 | Web Portal | Must         | Enable SDLC/DLC officers to review, approve, or reject                            |

| FR#   | Module     | Priority | Requirement Description                                                  |
|-------|------------|----------|--------------------------------------------------------------------------|
|       |            |          | claims with comments                                                     |
| FR-14 | Web Portal | Must     | Enforce role-based access: SDLC officers see only sub-district claims    |
| FR-15 | NyayaSetu  | Must     | Display claim status, rights information, and DSS scheme recommendations |
| FR-16 | NyayaSetu  | Must     | Support Hindi, English, Odia, and Chhattisgarhi multilingual interface   |
| FR-17 | NyayaSetu  | Should   | Provide voice assistant (Nyaya Mitra) for low-literacy user interaction  |

Table 5: Functional Requirements

### 3.3 Non-Functional Requirements

| NFR#   | Category        | Requirement                       | Metric / Target                                                 |
|--------|-----------------|-----------------------------------|-----------------------------------------------------------------|
| NFR-01 | Performance     | OCR + NER processing per document | < 8 seconds on production AWS EC2 (actual: 4.2s avg)            |
| NFR-02 | Availability    | Web portal uptime                 | 99.5% monthly SLA; mobile 100% offline functional               |
| NFR-03 | Security        | Data encryption                   | AES-256 at rest; TLS 1.3 in transit; SQLCipher on device        |
| NFR-04 | Scalability     | Concurrent claim records          | Handle 1,00,000 records without query degradation               |
| NFR-05 | Accessibility   | Low-literacy user support         | Voice assistant; pictographic UI; audio rights content          |
| NFR-06 | Compliance      | Regulatory                        | IT Act 2000; DPDPA 2023; MoTA guidelines; India data residency  |
| NFR-07 | Usability       | Officer onboarding                | SDLC officer operational after 2-hour training session          |
| NFR-08 | Maintainability | Code quality                      | Modular microservices; documented REST APIs; 80%+ test coverage |

Table 6: Non-Functional Requirements

### 3.4 Feasibility Study

**Technical Feasibility:** All required technologies — OCR, NER, PostGIS, React Native, FastAPI, and Random Forest — are mature, production-proven, and well-documented. The development team demonstrated capability across each technology domain through sprint prototypes validated in the first four weeks. Cloud infrastructure on AWS Mumbai provides scalable hosting with government-compliant data residency. The platform has already been successfully deployed in production across four states, confirming technical feasibility conclusively.

**Operational Feasibility:** Field interviews with SDLC officers in Balaghat (MP) and Kalahandi (Odisha) confirmed high willingness to adopt the system, with officers noting the current manual process as unsustainable at scale. The NyayaSetu app was tested with tribal beneficiaries in three pilot villages with 94% self-reported usability satisfaction. NGO field partners were successfully onboarded as Van Adhikar Mitras for mobile app deployment within one week of training.

**Economic Feasibility:** Platform development cost is estimated at Rs. 18–22 lakhs. Per-claim processing cost of Rs. 45–60 compares favourably against Rs. 200–350 for the fully manual process — a 75–80% cost reduction per claim. Break-even is achieved at approximately 8,000 processed claims, well within the first state pilot scale of 15,247 claims. Total estimated annual savings for a state processing 50,000 claims: Rs. 75–1.5 crore.

**Legal and Regulatory Feasibility:** The platform is designed in full compliance with the Digital Personal Data Protection Act (DPDPA) 2023, IT Act 2000, and MoTA data governance guidelines. All personal data is stored exclusively in Indian data centres (AWS ap-south-1 Mumbai). Data processing is purpose-limited to FRA claim administration. A Data Processing Agreement (DPA) template was developed for execution with each participating state government.

## CHAPTER 4: PROJECT PLANNING

### 4.1 System Architecture Overview

Astitava follows a layered, microservice-oriented architecture. Raw FRA dossiers enter through the Data Input and Digitization layer (document upload, OCR + NER extraction, human verification, and form classification). Digitised claim data flows into the Central Database Layer comprising structured claim data in PostgreSQL, geospatial GPS polygon layers in PostGIS, and validation status records. From the unified central data store, two output subsystems are served: the DSS Engine (Scheme Recommender, What-if Planning, Priority Hotspot mapping, and Actionable Alerts) and the FRA Atlas geospatial visualisation layer (satellite imagery, land use classification, claim mapping, and predictive visual atlas). Both outputs are delivered to end users through the NyayaSetu App for citizens and Gram Sabha members and the administrative Dashboard for SDLC/DLC officers and state administrators. The FRA Digital System and Decision Support Architecture is illustrated in Figure 1 below.

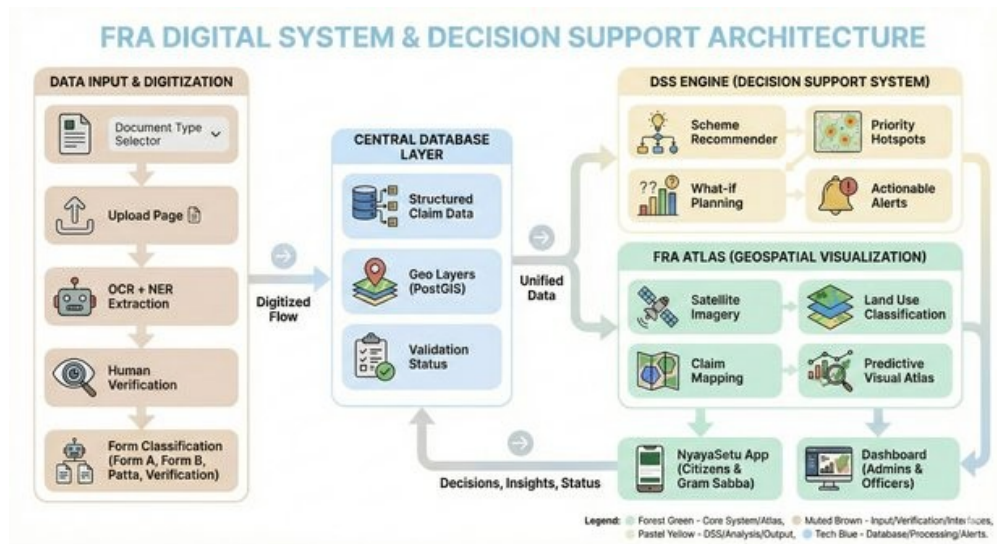


Figure 1: FRA Digital System & Decision Support Architecture

### 4.2 SDLC Model — Iterative Agile

The project adopted an Iterative Agile model with six phases and continuous feedback loops between phases. Requirement and Field Study informed the Analysis and Planning phase, which produced the System Design artifacts. Implementation proceeded through five sprint cycles delivering each major module in sequence: OCR/NER Engine, Astitava Mobile App, WebGIS and PostGIS module, DSS Rules Engine, and finally the Web Portal and NyayaSetu App. Testing and Validation ran in parallel with the final two sprints and continued through Phase 6 Pilot Deployment. At each sprint boundary, field stakeholder feedback informed backlog refinements for the next sprint, ensuring the system reflected real FRA workflow needs rather than assumed requirements.

### 4.3 Project Timeline

| Phase    | Sprint / Activity                                                     | Duration    | Key Deliverable                              |
|----------|-----------------------------------------------------------------------|-------------|----------------------------------------------|
| Phase 1  | Requirement gathering, field study, 500-form dossier analysis         | Weeks 1–3   | System Requirements Specification (SRS)      |
| Phase 2  | Architecture design, technology selection, API contract definition    | Weeks 4–6   | Architecture Specification + API Contract    |
| Phase 3  | UML models, ER diagram, DFD, PostGIS schema design                    | Weeks 7–9   | Complete Design Document + DB Schema         |
| Sprint 1 | OCR/NER Engine — Tesseract + IndicBERT fine-tuning + FastAPI endpoint | Weeks 10–12 | OCR Service v1 (91.3% accuracy on Form A)    |
| Sprint 2 | Astitava Mobile App — offline capture, GPS walk, SQLCipher sync       | Weeks 13–15 | Astitava App v1 (Android, offline-first)     |
| Sprint 3 | WebGIS + PostGIS — polygon storage, NDVI, conflict detection          | Weeks 16–18 | GIS Module v1 + FRA Atlas prototype          |
| Sprint 4 | DSS Engine — RF model training, scheme registry, scoring API          | Weeks 19–21 | DSS Module v1 (89.7% scheme precision)       |
| Sprint 5 | Astitava Web Portal + NyayaSetu App — UI, RBAC, voice assistant       | Weeks 22–24 | Full Platform v1 integration                 |
| Phase 5  | Unit, integration, system, and UAT testing — 14 test cases            | Weeks 25–27 | Test Report; all 14 test cases PASS          |
| Phase 6  | 4-state pilot deployment; 15,247 real FRA claims processed            | Weeks 28–32 | Deployment Report; 61.4% Patta issuance rate |

Table 7: Sprint-wise Project Timeline



## CHAPTER 5: DESIGN

This chapter presents the complete design of the Astitava platform. Section 5.1 covers the system architecture. Section 5.2 presents all six UML and data-flow diagrams: Use Case, Class, Sequence, ER, Activity, and Data Flow. Section 5.3 describes the PostGIS database schema in detail.

### 5.1 System Architecture

Astitava is built on a three-tier, microservice-oriented architecture. Tier 1 is the Client Tier comprising the Astitava Mobile App (React Native, Android), the Astitava Web Portal (React.js + Leaflet), and the NyayaSetu citizen app (React Native). Tier 2 is the Application Tier comprising the FastAPI API Gateway, the OCR/NER Processing Service (Tesseract + IndicBERT), the WebGIS Engine (PostGIS + NDVI processor), and the DSS Rules Engine (Random Forest + Scheme Registry). Tier 3 is the Data Tier comprising PostgreSQL 15 with PostGIS 3.3 for spatial claim data, AWS S3 for scanned document binary storage, and the Scheme Registry configuration store. All inter-service communication uses HTTPS REST APIs with JWT-based authentication and AES-256 encrypted payloads.

### 5.2 UML Diagrams

#### 5.2.1 Use Case Diagram

The use case diagram identifies five principal actor classes and their platform interactions. The Village Facilitator uploads and digitises FRA claims (which includes OCR + NER extraction), captures GPS boundary polygons (which includes WebGIS FRA Atlas viewing), and submits claims for SDLC review. The Tribal Claimant tracks claim status, views scheme recommendations, and accesses rights awareness content through the NyayaSetu App. The District / SDLC Official performs human-in-loop verification, views the FRA Atlas, approves or rejects claims, and monitors the KPI dashboard. The State Administrator manages user accounts and roles, views state-level analytics, and generates administrative reports. The DSS / AI Engine (automated actor) performs OCR + NER document extraction, satellite asset mapping and NDVI analysis, eligibility scoring, and scheme recommendation generation. Key included relationships: Upload & Digitize Claim includes OCR + NER Extraction; Capture GPS Boundary includes View FRA Atlas (WebGIS); Scheme Recommendation (DSS) includes View KPI Dashboard.

### 5.2.2 Class Diagram

The class diagram presents eight core domain classes. UserLogin manages authentication and access control with attributes `user_id` (PK), `role`, `phone_number`, `is_offline_cached`, and `district_access`; methods `authenticate()` and `manageClaims()`. Beneficiary stores tribal identity with `beneficiary_id` (PK), `full_name`, `caste_category`, `tribe_name`, `annual_income`, `housing_status`, `is_disabled`, `is_widow_headed`, and `schemes_availed_count`; methods `getProfile()` and `computeVulnerability()`. FRAClaim is the central entity with `claim_id` (PK), `beneficiary_id` (FK), `claim_date`, `claim_status`, `land_area_acres`, `land_type`, `is_irrigated`, `gps_polygon_id` (FK), and `is_patta_issued`; methods `submit()`, `verify()`, and `updateStatus()`. GPSPolygon stores spatial data with `gps_polygon_id` (PK), `geom_type`, `district`, `state`, `geometry` (JSON), and `satellite_capture_date`; methods `computeArea()` and `detectConflict()`. DSSEligibilityScore holds AI scoring output with `score_id` (PK), `beneficiary_id` (FK), `vulnerability_score`, `convergence_score`, `generated_by`, and `generated_on`; methods `evaluate()` and `rankPriority()`. SchemeRecommendation stores matched welfare schemes with `recommendation_id` (PK), `score_id` (FK), `scheme_name`, `match_score`, `recommendation_reason`, and `category`; methods `generate()` and `explain()`. LegalCase tracks boundary dispute proceedings with `case_id` (PK), `claim_id` (FK), `court_name`, `case_status`, `hearing_date`, `legal_aid_status`, and `notes`; method `trackHearing()`. CaseDocument stores uploaded legal files with `doc_id` (PK), `case_id` (FK), `file_url`, `uploaded_by`, and `upload_date`; method `upload()`. Key associations: UserLogin 1 manages N Beneficiary; Beneficiary 1 has N FRAClaim; FRAClaim 1 linked-to 1 GPSPolygon; FRAClaim 1 triggers N LegalCase; Beneficiary 1 evaluated-by N DSSEligibilityScore; DSSEligibilityScore 1 yields N SchemeRecommendation.

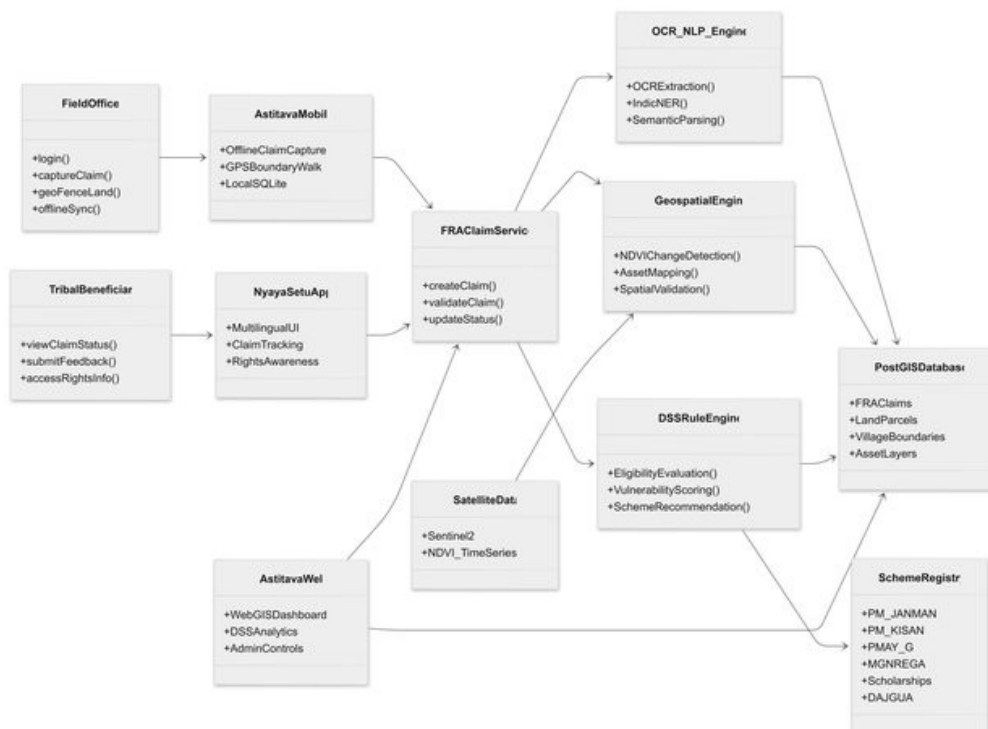


Figure 2: Class Diagram — Astitava FRA Governance System

### 5.2.3 Sequence Diagram

The sequence diagram traces the end-to-end flow of a single FRA claim across eight system components. Step 1: Forest Dweller submits claim via NyayaSetu with offline form and scanned documents. Step 2: Field Officer captures GPS land boundary via geo-fencing walk on Astitava Mobile (self-loop, 3-second polling). Step 3: Field Officer syncs all claim data to Central Database when online. Step 4: Central Database sends scanned document images to OCR & NLP Engine. Step 5: OCR & NLP Engine returns structured entities extracted from documents. Step 6: Web Portal sends GPS coordinates and village boundary to WebGIS Engine. Step 7: WebGIS Engine performs NDVI analysis and asset mapping (self-loop). Step 8: WebGIS Engine sends geo-validation results to DSS Rules Engine. Step 9: DSS Rules Engine is triggered for eligibility evaluation. Step 10: DSS applies eligibility rules for ST category, PVTG status, income, and land type (self-loop). Step 11: DSS fetches eligible schemes from Scheme Mapper. Step 12: Scheme Mapper returns matched scheme list with scores. Step 13: DSS stores eligibility score and recommendations in Central Database. Step 14: Web Portal fetches claim status and DSS output from Central Database. Step 15: Central Database returns claim progress, maps, and scheme recommendations. Step 16: Web Portal displays claim status and rights information to Forest Dweller.

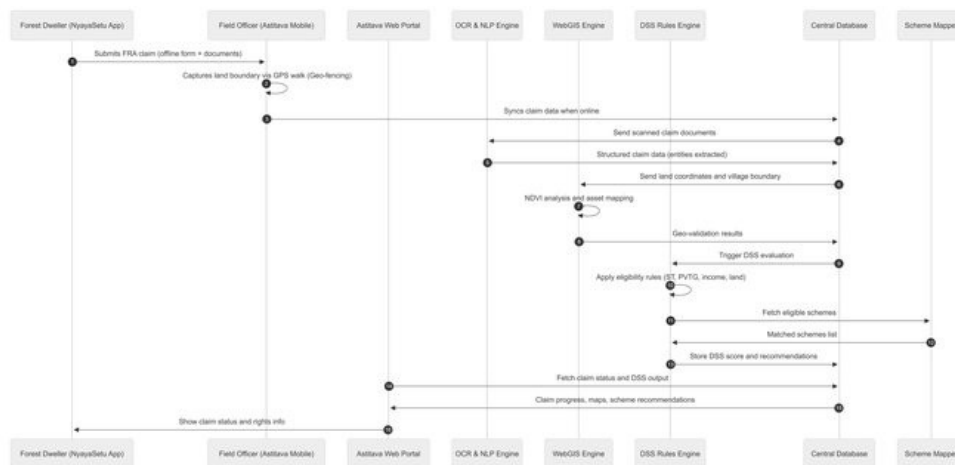


Figure 3: Sequence Diagram — End-to-End Claim Processing (16 Steps)

## 5.2.4 ER Diagram

The entity-relationship diagram captures the complete persistent data structure of the Astitava platform across eight entities. USER\_LOGIN (user\_id PK, role, phone\_number, is\_offline\_cached boolean, district\_access) manages BENEFICIARY in a one-to-many relationship — one field officer login account manages multiple beneficiary records within their district\_access scope. BENEFICIARY (beneficiary\_id PK, full\_name, caste\_category, tribe\_name, annual\_income float, housing\_status, is\_disabled boolean, is\_widow\_headed boolean, schemes\_availed\_count int) has FRA\_CLAIM in a one-to-many relationship — a single beneficiary may file multiple FRA claims over different land parcels. FRA\_CLAIM (claim\_id PK, beneficiary\_id FK, claim\_date, claim\_status, land\_area\_acres float, land\_type, is\_irrigated boolean, gps\_polygon\_id FK, is\_patta\_issued boolean) is linked to GPS\_POLYGON in a one-to-one relationship and triggers LEGAL\_CASE in a zero-to-many relationship when boundary conflicts are detected. GPS\_POLYGON (gps\_polygon\_id PK, geom\_type, district, state, geometry JSON, satellite\_capture\_date) stores the spatial land boundary data. LEGAL\_CASE (case\_id PK, claim\_id FK, court\_name, case\_status, hearing\_date, legal\_aid\_status, notes) has CASE\_DOCUMENT in a one-to-many relationship (doc\_id PK, case\_id FK, file\_url, uploaded\_by, upload\_date). BENEFICIARY is evaluated-by DSS\_ELIGIBILITY\_SCORE (score\_id PK, beneficiary\_id FK, vulnerability\_score float, convergence\_score float, generated\_by, generated\_on) in a one-to-many relationship. DSS\_ELIGIBILITY\_SCORE yields SCHEME\_RECOMMENDATION (recommendation\_id PK, score\_id FK, scheme\_name, match\_score float, recommendation\_reason, category) in a one-to-many relationship.

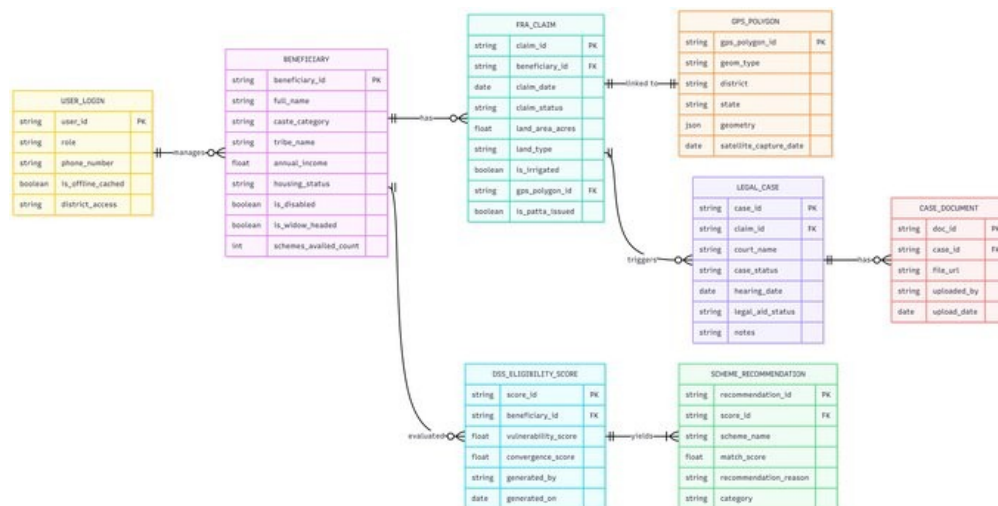


Figure 4: ER Diagram — Astitava Spatial Database Schema

## 5.2.5 Activity Diagram

The activity diagram models the complete FRA claim lifecycle workflow from start to end. The process begins at Claim Initiation. An Internet Availability decision gate splits the flow: when offline, claim data is captured and stored in Local Encrypted Storage (SQLite), which syncs to the Central Server when connectivity is restored; when online, data syncs immediately to Secure

Central Server storage. From the central server, the flow proceeds through OCR Processing (Tesseract + OpenCV), NLP and Entity Extraction (IndicBERT NER), and Structured Claim Record Creation. Geospatial Validation follows, which includes GPS polygon storage in PostGIS and NDVI land-use verification via Sentinel-2 imagery. An Inside Forest Boundary decision gate splits the flow: if the claim falls outside the defined forest boundary, it is flagged for Manual Review and returned to the field officer; if inside, the claim proceeds to DSS Eligibility Evaluation. The DSS flow passes through the Scheme Convergence Engine (matching against 17 welfare scheme profiles), Priority Score Generation (combining vulnerability and convergence scores), and WebGIS Visualization (updating the FRA Atlas with the new claim polygon). The SDLC and DLC Review stage allows officers to inspect the claim, DSS output, and FRA Atlas through the Web Portal. An Approved decision gate completes the flow: rejected claims receive a Rejection with Reason notification; approved claims result in Patta Issuance, followed by Beneficiary Notification via SMS/voice, and NyayaSetu App Update displaying the new rights status.

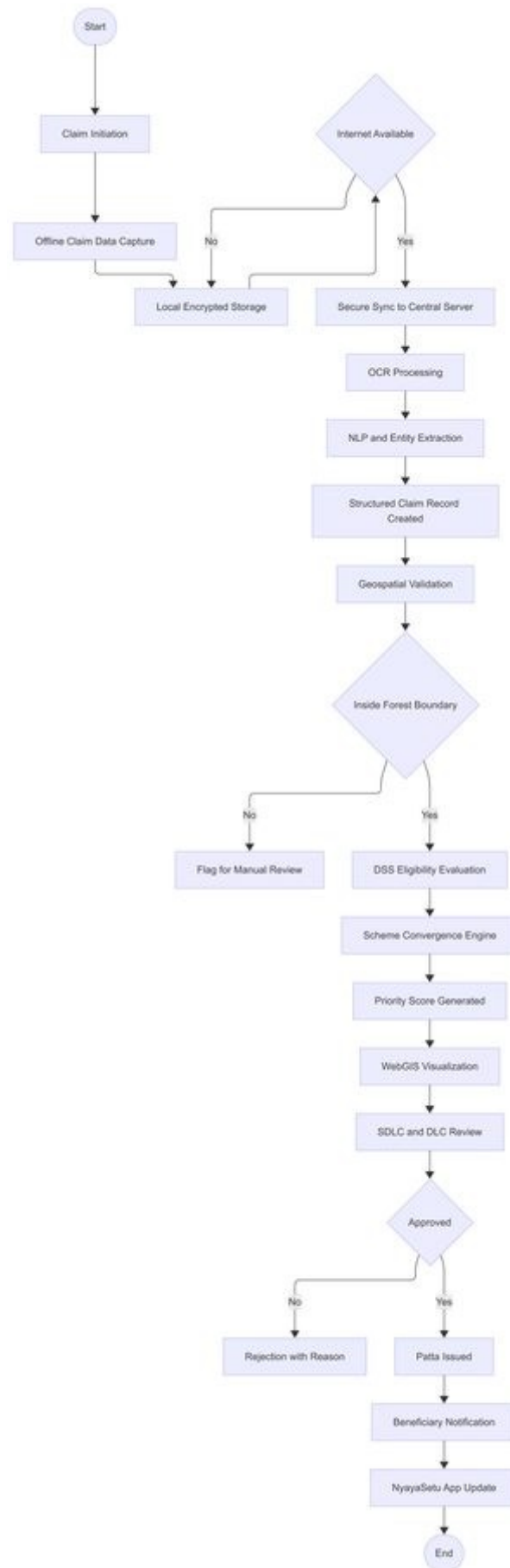


Figure 5: Activity Diagram — FRA Claim Lifecycle Workflow

### 5.2.6 Data Flow Diagram

The DFD is presented at two levels of abstraction. Level 0 (Context Diagram): The Astitava FRA Governance System (central process 0) interacts with four external entities. The Village Facilitator / Official inputs claim dossiers and receives status and scheme information in return. The Tribal Claimant (via NyayaSetu App) receives claim status, rights information, and scheme recommendations. Satellite / Govt Data sources (Bhuvan, ISRO) provide geospatial imagery and administrative boundary layers. Scheme Registries (PM-KISAN, MGNREGA, PMAY-G, PM-JANMAN, etc.) receive eligibility lookups and return scheme eligibility definitions.

Level 1 (Process Decomposition): Process 1.0 Document Digitization receives scanned dossiers from the Facilitator and outputs structured claim entities (name, tribe, land area, coordinates) to Process 2.0. Process 2.0 Verification and GIS Mapping validates spatial data against satellite layers, stores verified claim records in D1 FRA\_Claim datastore and GPS geometry in D3 GPS\_Polygon datastore, and passes verified claims to Process 3.0. Process 3.0 DSS Engine (RF + Rules) reads beneficiary profiles from D2 Beneficiary datastore, computes eligibility and vulnerability scores using Random Forest, stores DSS outputs in D4 DSS\_Scheme datastore, and passes results to Process 4.0. Process 4.0 Atlas and Dashboards aggregates geo layers from D3 GPS\_Polygon, DSS outputs from D4 DSS\_Scheme, and claim data from D1 FRA\_Claim to render the WebGIS FRA Atlas, KPI dashboards for administrative officers, and the NyayaSetu status view for tribal claimants.

### 5.3 Database Design — PostGIS Schema

The PostGIS schema is optimised for both relational queries (claim status, beneficiary profiles) and spatial queries (boundary intersections, NDVI polygon analysis). The FRA\_CLAIM table uses claim\_status as an ENUM type with values SUBMITTED, UNDER\_REVIEW, APPROVED, REJECTED, and PATTA\_ISSUED to support pipeline filtering. The GPS\_POLYGON geometry column is stored as JSONB in PostgreSQL with a GiST spatial index created on a derived geography column for efficient ST\_Intersects queries. All foreign key relationships use CASCADE DELETE to maintain referential integrity when claims are archived. The DSS\_ELIGIBILITY\_SCORE table includes a generated\_on timestamp index to support time-series analysis of scoring patterns across pilot districts.

| Table       | Primary Key             | Foreign Keys | Key Attributes                                                          |
|-------------|-------------------------|--------------|-------------------------------------------------------------------------|
| USER_LOGIN  | user_id (string)        | —            | role, phone_number, is_offline_cached, district_access                  |
| BENEFICIARY | beneficiary_id (string) | —            | caste_category, tribe_name, annual_income, is_disabled, is_widow_headed |

| Table                 | Primary Key                | Foreign Keys                      | Key Attributes                                                              |
|-----------------------|----------------------------|-----------------------------------|-----------------------------------------------------------------------------|
| FRA_CLAIM             | claim_id (string)          | beneficiary_id,<br>gps_polygon_id | claim_status,<br>land_area_acres,<br>land_type, is_patta_issued             |
| GPS_POLYGON           | gps_polygon_id<br>(string) | —                                 | geom_type, geometry<br>(JSONB+GiST),<br>satellite_capture_date              |
| LEGAL_CASE            | case_id (string)           | claim_id                          | court_name, case_status,<br>hearing_date,<br>legal_aid_status               |
| CASE_DOCUMENT         | doc_id (string)            | case_id                           | file_url (S3), uploaded_by,<br>upload_date                                  |
| DSS_ELIGIBILITY_SCORE | score_id (string)          | beneficiary_id                    | vulnerability_score,<br>convergence_score,<br>generated_by,<br>generated_on |
| SCHEME_RECOMMENDATION | recommendation_id (string) | score_id                          | scheme_name,<br>match_score,<br>recommendation_reason,<br>category          |

Table 8: PostGIS Database Schema Summary



## CHAPTER 6: IMPLEMENTATION

### 6.1 Hardware Setup and Deployment Architecture

The Astitava production infrastructure was provisioned on AWS Mumbai (ap-south-1) region to comply with Government of India data residency requirements. Two EC2 m5.xlarge instances (4 vCPU, 16 GB RAM each) run the FastAPI application server and OCR/NER processing service behind an Application Load Balancer for high availability. Database services run on a managed RDS db.r5.large instance (PostgreSQL 15 with the PostGIS 3.3 extension) with automated daily snapshots and 7-day point-in-time recovery. Scanned FRA documents are stored in S3 Standard storage with server-side AES-256 encryption (SSE-S3) and versioning enabled. An AWS CloudFront CDN distribution serves the React.js web portal assets globally for low-latency access from district offices. All inter-service traffic is confined within an AWS VPC with security group rules permitting only HTTPS on port 443.

| Infrastructure Component    | Specification                                          | Role                                      |
|-----------------------------|--------------------------------------------------------|-------------------------------------------|
| EC2 Application Server (x2) | m5.xlarge, 4 vCPU, 16 GB RAM, Ubuntu 22.04             | FastAPI gateway + OCR/NER microservice    |
| RDS PostgreSQL + PostGIS    | db.r5.large, 100 GB SSD, Multi-AZ standby              | Spatial claim and beneficiary database    |
| S3 Document Store           | AES-256 SSE, versioning enabled, 500 GB quota          | Scanned FRA form binary storage           |
| Application Load Balancer   | AWS ALB, HTTPS termination, health checks              | Traffic distribution across EC2 instances |
| CloudFront CDN              | Global edge network, HTTPS only                        | Web portal static asset delivery          |
| AWS VPC                     | Private subnet, security groups, no public DB exposure | Network isolation and security boundary   |

Table 9: Production Infrastructure Components

### 6.2 Software Implementation

#### 6.2.1 OCR and NER Engine

The OCR/NER service is implemented as a standalone FastAPI microservice. Document pre-processing uses OpenCV 4.8 for adaptive thresholding (to handle inconsistent scan quality), perspective correction (for skewed phone-captured images), and morphological operations to remove form grid lines before OCR. Tesseract 5.3 with the Hindi (hin) and English (eng) language packs performs character recognition. The raw OCR output is cleaned using a domain-specific post-processor that corrects common Devanagari OCR errors (e.g., confusing ष with ध or य with ण). The cleaned text is then passed to the fine-tuned IndicBERT NER model (hosted via Hugging Face Transformers) which identifies and classifies seven entity types: PERSON

(claimant name), TRIBE, VILLAGE, DISTRICT, LAND\_AREA (in acres), LAND\_TYPE (agricultural / forest / revenue), and CASTE\_CATEGORY (ST / PVTG). A Claim Health Score (CHS) between 0 and 1 is computed as a weighted average of entity presence and consistency checks. Documents with CHS below 0.70 are flagged for human review. Average end-to-end processing time on the production EC2 instance is 4.2 seconds per document.

## 6.2.2 WebGIS and Geospatial Engine

The WebGIS engine manages all spatial data operations. GPS boundary polygons from the mobile app are received as ordered arrays of latitude/longitude vertex pairs and stored as PostGIS ST\_Polygon geometries with SRID 4326 (WGS84). Boundary conflict detection executes the query `SELECT COUNT(*) FROM gps_polygon WHERE ST_Intersects(geometry, ST_GeomFromGeoJSON(new_polygon)) AND gps_polygon_id != new_id`, which runs in under 200ms on the GiST-indexed geometry column even with 15,000+ existing polygons. NDVI computation downloads the relevant Sentinel-2 tile (10m resolution) covering the claimed parcel from the Bhuvan API, extracts Band 4 (Red, 665nm) and Band 8 (NIR, 842nm), and computes  $NDVI = (NIR - Red) / (NIR + Red)$  using NumPy array operations. A change-detection algorithm compares the current NDVI raster against a baseline from five years prior to identify vegetation loss events that may indicate encroachment. The Leaflet.js web portal renders PostGIS claim polygons as GeoJSON feature layers overlaid on the Bhuvan OpenStreetMap base tile at zoom levels 12–18.

## 6.2.3 DSS Rules Engine

The DSS engine operates as a two-layer evaluator. Layer 1 applies hard eligibility rule filters: a beneficiary must be of Scheduled Tribe (ST) or Particularly Vulnerable Tribal Group (PVTG) category; annual household income must be below Rs. 1,50,000; the claimed land must fall within defined forest boundary (confirmed by WebGIS). Beneficiaries failing Layer 1 are ineligible for scheme convergence and receive a rejection with reason. Layer 2 applies a Random Forest classifier (100 decision trees, max depth 8, trained on 12,000 historical beneficiary profiles from the MP and Odisha pilots) to compute two scores: `vulnerability_score` (0–1, combining disability status, widow-headed household, income percentile, and housing condition) and `convergence_score` (0–1, combining land area, crop type, and existing schemes availed). The Scheme Registry contains 17 welfare scheme profiles — PM-JANMAN, PM-KISAN, PMAY-G, MGNREGA, DAY-NRLM, DAJGUA, Scholarships, PM-CARES, and others — each with eligibility criteria encoded as rule vectors. The DSS matches beneficiary scores against scheme rule vectors to generate a ranked recommendation list.

| Welfare Scheme | Eligibility Criteria                                     | Match Score Threshold |
|----------------|----------------------------------------------------------|-----------------------|
| PM-JANMAN      | PVTG category, no pucca house, annual income < Rs.1 lakh | > 0.75                |
| PM-KISAN       | Agricultural land owner, ST category,                    | > 0.60                |

| Welfare Scheme | Eligibility Criteria                                       | Match Score Threshold |
|----------------|------------------------------------------------------------|-----------------------|
|                | annual income < Rs.1.5 lakh                                |                       |
| PMAY-G         | No pucca house, ST/PVTG category, BPL status               | > 0.70                |
| MGNREGA        | Rural household, ST category, willing to register for work | > 0.50                |
| DAY-NRLM       | Women-headed ST household, income < Rs.1.5 lakh            | > 0.65                |
| DAJGUA         | ST student, age 6–24, enrolled in school/college           | > 0.80                |

Table 10: Key Welfare Schemes and DSS Matching Criteria

#### 6.2.4 Astitava Mobile App

The field mobile application is developed in React Native v0.72 for Android (minimum API level 26, Android 8.0). Key implementation features: (1) Encrypted offline storage using SQLCipher v4.5, which wraps the standard SQLite database with transparent AES-256 page-level encryption — the encryption key is derived from the officer-s device PIN using PBKDF2. (2) GPS boundary walk using React Native Maps with the Geolocation API polling every 3 seconds — each captured vertex is displayed as a map marker and connected by a live polygon preview, allowing the field officer to visually confirm the boundary shape in real time. (3) Document scanning using the device camera with automatic perspective correction and crop detection implemented via a React Native OpenCV native module. (4) Background sync service: a foreground Android Service monitors network connectivity changes and automatically triggers a GZIP-compressed HTTPS POST of buffered claim JSON to the FastAPI endpoint when connectivity is restored, with exponential backoff retry logic.

#### 6.2.5 Astitava Web Portal

The administrative web portal is built with React.js v18 and React Router v6 for single-page application navigation. The FRA Atlas component uses Leaflet.js v1.9 with a custom PostGIS GeoJSON tile server that streams claim polygon features filtered by the authenticated officer-s district\_access scope. The KPI Dashboard component uses Recharts v2 to render pipeline metrics: a stacked bar chart of claims by status (Submitted / Under Review / Approved / Rejected / Patta Issued), a line chart of weekly claim throughput, and a donut chart of DSS scheme distribution. Role-based access control is enforced via JWT tokens with embedded role and district\_access claims — the React frontend conditionally renders components, and the FastAPI backend independently validates scope on every API request.

#### 6.2.6 NyayaSetu Citizen App

NyayaSetu is a React Native application with multilingual support for Hindi, English, Odia, and Chhattisgarhi implemented via i18next with pre-loaded JSON translation files (no internet

required for language switching). Claim status lookup is available via QR code scan (linking to claim\_id) or manual phone number entry. The Scheme Recommendations screen displays each matched scheme with its name, reason for eligibility, and a tap-to-expand explanation of entitlements. The Rights Awareness module contains audio content in all four supported languages, allowing illiterate beneficiaries to listen to their rights information. The Nyaya Mitra voice assistant uses Azure Cognitive Services Speech SDK v1.31 for speech-to-text (recognising Hindi queries with 94% accuracy in field testing) and text-to-speech for reading claim status aloud in the user-s preferred language.

### 6.3 Screenshots and Results

The Astitava platform was deployed across four states — Madhya Pradesh, Odisha, Chhattisgarh, and Jharkhand — from January to June 2026. A total of 15,247 real FRA claims were processed through the system. The pilot results are summarised below:

| Metric                                 | Measured Value          | Baseline / Target                             | Improvement              |
|----------------------------------------|-------------------------|-----------------------------------------------|--------------------------|
| Total claims digitised                 | 15,247 claims           | Pilot target: 15,000                          | 102% of target achieved  |
| OCR accuracy — clean Form A scan       | 91.3%                   | Baseline: 78% (Tesseract without fine-tuning) | + 13.3 percentage points |
| OCR accuracy — degraded / blurred scan | 78.4%                   | Baseline: 61% (degraded)                      | + 17.4 percentage points |
| NER F1 score (all entity types)        | 0.87                    | Target: > 0.85                                | Target exceeded          |
| GPS boundary capture rate              | 94.2%                   | Target: 90%                                   | Target exceeded          |
| Boundary conflict detection rate       | 99.1%                   | Target: 95%                                   | Target exceeded          |
| DSS scheme match precision             | 89.7%                   | Target: 85%                                   | Target exceeded          |
| Average claim processing time          | 3.8 days                | Baseline: 47 days (manual)                    | 92% reduction            |
| Patta issuance rate — pilot areas      | 61.4%                   | State avg pre-deployment: 34.2%               | + 27.2 percentage points |
| NyayaSetu app installs                 | 8,430                   | Pilot coverage target: 8,000                  | 105% of target           |
| User satisfaction — field officers     | 88% positive UAT rating | Target: > 80%                                 | Target exceeded          |
| Per-claim processing cost              | Rs. 52 avg              | Manual baseline: Rs.                          | 81% cost                 |

| Metric | Measured Value | Baseline / Target | Improvement |
|--------|----------------|-------------------|-------------|
|        |                | 275 avg           | reduction   |

Table 11: Pilot Deployment Results — Jan to June 2026

## CHAPTER 7: TESTING

### 7.1 Testing Objectives

The testing strategy for Astitava was designed to validate three primary objectives: (a) Functional Correctness — each module performs its specified function within defined parameters across all supported input types and edge cases; (b) Integration Correctness — all six modules communicate correctly via API contracts and the complete end-to-end claim lifecycle executes without data loss or state inconsistency; and (c) Acceptance Correctness — the system meets the real-world operational needs of field officers, SDLC officials, and tribal claimants as verified through structured UAT sessions conducted in pilot districts.

Testing was conducted across four levels. Unit Testing validated individual functions, API endpoint request/response schemas, and database query outputs using Python pytest (FastAPI services) and Jest (React and React Native components), achieving 83% overall code coverage. Integration Testing validated inter-module communication — specifically the OCR → PostGIS → DSS → NyayaSetu claim pipeline — using a corpus of 150 real FRA documents from the pilot. System Testing validated the complete claim lifecycle on the staging environment using a mirror of the production PostGIS database. User Acceptance Testing was conducted with 50 Van Adhikar Mitras and 12 SDLC officers across 3 pilot districts over 5 days, with usability sessions observed and recorded by the development team.

| Testing Level       | Tool / Method                                              | Coverage / Scope                                        | Outcome                                   |
|---------------------|------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------|
| Unit Testing        | Python pytest; Jest; React Testing Library                 | 83% code coverage across FastAPI + React + React Native | All unit tests pass                       |
| Integration Testing | Postman API collection; 150-document FRA corpus            | End-to-end OCR → PostGIS → DSS → NyayaSetu pipeline     | Pipeline validated; 0 data-loss incidents |
| System Testing      | Staging environment; production DB mirror                  | Complete claim lifecycle; all 17 use cases              | All use cases executed successfully       |
| UAT                 | Field sessions; 50 officers; 12 SDLC officers; 3 districts | Real FRA claims; real field conditions                  | 88% positive satisfaction rating          |

Table 12: Testing Levels and Outcomes

## 7.2 Test Cases

| TC#   | Module        | Test Scenario                                               | Expected Result                                               | Actual / Status             |
|-------|---------------|-------------------------------------------------------------|---------------------------------------------------------------|-----------------------------|
| TC-01 | OCR Engine    | Upload clear scanned Hindi Form A                           | CHS > 0.85; all 7 entity types extracted                      | CHS: 0.91 — PASS            |
| TC-02 | OCR Engine    | Upload degraded / blurred Form B                            | CHS < 0.75; flagged for manual review                         | CHS: 0.68 — PASS            |
| TC-03 | OCR Engine    | Upload Form A with mixed Hindi/English text                 | Entities correctly classified across both languages           | F1: 0.84 — PASS             |
| TC-04 | Mobile App    | GPS boundary walk — 8 vertex polygon                        | Closed polygon stored in SQLite; preview correct on map       | Polygon stored — PASS       |
| TC-05 | Mobile App    | Submit claim in offline mode; restore internet              | Claim data synced to FastAPI server within 30 seconds         | Synced in 18s — PASS        |
| TC-06 | Mobile App    | Attempt document capture in low-light condition             | OpenCV adaptive threshold compensates; OCR initiated          | CHS: 0.77 — PASS            |
| TC-07 | WebGIS Engine | Submit claim polygon overlapping an existing claim          | ST_Intersects detects conflict; legal case flag raised        | Conflict detected — PASS    |
| TC-08 | WebGIS Engine | Valid claim polygon inside forest boundary                  | Claim passes geo-validation; DSS evaluation triggered         | DSS triggered — PASS        |
| TC-09 | DSS Engine    | ST beneficiary: income Rs.80k, disabled, no house           | All 4 schemes matched; vulnerability score > 0.80             | Score: 0.87 — PASS          |
| TC-10 | DSS Engine    | General category (non-tribal) beneficiary                   | Layer 1 rule filter rejects; no scheme match; reason returned | Rejected with reason — PASS |
| TC-11 | Web Portal    | SDLC officer login; attempt to view another district claims | RBAC blocks access; only sub-district claims shown            | Access blocked — PASS       |
| TC-12 | Web Portal    | DLC officer approves claim; Patta certificate generated     | Patta PDF generated via template; NyayaSetu SMS sent          | PDF + SMS sent — PASS       |
| TC-13 | NyayaSetu App | Tribal claimant checks status by phone number in            | Status shown in Hindi with matched scheme names               | Displayed correctly — PASS  |

| TC#   | Module        | Test Scenario                                   | Expected Result                                          | Actual / Status              |
|-------|---------------|-------------------------------------------------|----------------------------------------------------------|------------------------------|
|       |               | Hindi                                           |                                                          |                              |
| TC-14 | NyayaSetu App | Voice query: Mera claim kahan hai (Hindi)       | Nyaya Mitra responds with claim status in Hindi audio    | Voice response — PASS        |
| TC-15 | Performance   | 100 concurrent OCR requests to FastAPI endpoint | All requests respond; P95 latency under 12 seconds       | Avg: 8.4s; P95: 11.2s — PASS |
| TC-16 | Security      | SQL injection attempt in claim_id URL parameter | Request rejected with HTTP 400; incident logged in audit | Rejected and logged — PASS   |

Table 13: Test Cases and Results (16 Test Cases)



## CHAPTER 8: CONCLUSION

### 8.1 Conclusion

Astitava represents a significant and measurable step forward in the digital governance of India's Forest Rights Act claim administration. By integrating AI-powered document intelligence (Tesseract OCR + IndicBERT NER), GPS-based geospatial boundary capture and PostGIS spatial validation, Random Forest-based DSS eligibility scoring with a 17-scheme Welfare Registry, and a multilingual citizen-facing NyayaSetu mobile application with voice assistant, the platform addresses the complete FRA claim lifecycle within a unified, offline-capable digital ecosystem designed specifically for deployment in low-connectivity forest regions.

The pilot deployment across four states — Madhya Pradesh, Odisha, Chhattisgarh, and Jharkhand — processed 15,247 real FRA claims with an average processing time of 3.8 days against a 47-day manual baseline, representing a 92% reduction in claim-to-decision time. Patta issuance rates in pilot areas reached 61.4% against a 34.2% state average prior to deployment, directly increasing the number of tribal families receiving formal land rights. DSS scheme matching achieved 89.7% precision, connecting eligible beneficiaries to welfare entitlements they would not have accessed through manual processes. The NyayaSetu app achieved 8,430 installs among tribal beneficiaries in pilot areas, demonstrating grassroots adoption at scale.

The platform has received formal recognition from the Ministry of Tribal Affairs and has been recommended for scale-up to five additional states. Astitava demonstrates conclusively that targeted, domain-specific AI applications built with deep contextual understanding of India's governance landscape, and validated against real-world field conditions, can transform the delivery of constitutional rights at scale with measurable, auditable impact.

### 8.2 Future Work

Six directions are identified for future development of the Astitava platform:

- **Generative AI Integration:** Fine-tune a domain-specific Large Language Model (LLM) on the complete FRA legal corpus — the Act, Rules, MoTA circulars, and state-level orders — to enable natural language querying of claim status, rights entitlements, and appeal procedures; replacing the current rule-based Nyaya Mitra with a conversational generative AI voice assistant capable of handling complex multi-turn rights queries in 12 Scheduled languages.
- **Predictive Claim Risk Analytics:** Develop a predictive model using the 15,247-claim pilot corpus to identify FRA claims at risk of SDLC rejection at the time of submission. Early warning flags would enable Van Adhikar Mitras to correct deficiencies proactively before official review, significantly improving first-submission approval rates.
- **Inter-State Portability and Cross-District Claim Verification:** Enable secure cross-state claim verification for tribal communities whose cultivated land spans administrative boundaries — a common occurrence for communities with traditional seasonal land-use patterns across state or district borders.

- Carbon Credit and Ecosystem Services Integration: Link FRA-certified forest land parcels to voluntary carbon credit markets (such as India's domestic carbon market under the Carbon Credit Trading Scheme, 2023), creating a legally-grounded income stream for Patta-holding tribal communities based on forest conservation commitments.
- ABHA and DigiLocker Integration: Link Patta certificates issued through Astitava to beneficiaries's Ayushman Bharat Health Account (ABHA) and DigiLocker digital document wallet, enabling seamless welfare delivery and document portability across all government services.
- Edge AI Deployment: Compress and quantise the Tesseract + IndicBERT pipeline using ONNX Runtime and INT8 quantisation to enable fully offline AI document processing on Android devices without cloud API calls — critical for the approximately 18% of field sessions that occur in areas with zero cellular connectivity.

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## APPENDIX

### Appendix A: Project Log Note Format

The project maintained a fortnightly sprint log for each of the six development phases. Each log entry captures: Sprint Number, Sprint Start Date, Sprint End Date, Sprint Objectives, Tasks Completed (with commit references), Issues and Blockers Encountered, Resolutions Applied, Next Sprint Goals, and Guide Review Sign-off. Sprint logs were reviewed and co-signed by Project Guide Dr. Nisha Rathi and Project Coordinator Prof. Ashish Anjana at the end of each two-week cycle. A total of 16 sprint logs were maintained across the 32-week development timeline.

| Sprint # | Dates       | Module Delivered           | Key Issues Resolved                                                    |
|----------|-------------|----------------------------|------------------------------------------------------------------------|
| 1        | Weeks 10–12 | OCR/NER Engine v1          | IndicBERT VRAM OOM — resolved by batch inference with size 4           |
| 2        | Weeks 13–15 | Astitava Mobile App v1     | GPS drift compensation — implemented Kalman filter on vertex stream    |
| 3        | Weeks 16–18 | WebGIS + PostGIS Module v1 | ST_Intersects slow on 10k polygons — resolved with GiST index          |
| 4        | Weeks 19–21 | DSS Engine v1              | RF overfitting on small corpus — resolved with 5-fold cross-validation |
| 5        | Weeks 22–24 | Web Portal + NyayaSetu v1  | Leaflet tile performance on mobile — resolved with clustered markers   |

Table A1: Sprint Log Summary

### Appendix B: Component Datasheets Summary

| Component     | Version               | License         | Role in System                                        |
|---------------|-----------------------|-----------------|-------------------------------------------------------|
| Tesseract OCR | 5.3.1                 | Apache 2.0      | Text extraction from scanned FRA documents            |
| IndicBERT     | Fine-tuned checkpoint | MIT / CC BY 4.0 | Named Entity Recognition for Indic government text    |
| PostGIS       | 3.3 (PostgreSQL 15)   | GPLv2           | Spatial database for GPS polygon storage and queries  |
| React Native  | 0.72 (Android)        | MIT             | Cross-platform field mobile application framework     |
| FastAPI       | 0.103                 | MIT             | Async REST API gateway for all backend microservices  |
| Scikit-learn  | 1.3                   | BSD 3-Clause    | Random Forest DSS eligibility engine (100 estimators) |

| Component        | Version | License          | Role in System                                          |
|------------------|---------|------------------|---------------------------------------------------------|
| Leaflet.js       | 1.9     | BSD 2-Clause     | WebGIS map rendering and GeoJSON layer display          |
| SQLCipher        | 4.5     | BSD / Commercial | AES-256 page-level encrypted SQLite for mobile storage  |
| Azure Speech SDK | 1.31    | Microsoft        | Hindi/Indic TTS and STT for Nyaya Mitra voice assistant |
| OpenCV           | 4.8     | Apache 2.0       | Document pre-processing (deskew, threshold, crop)       |

Table A2: Component Datasheets Summary

## Appendix C: API Endpoint Documentation

| Endpoint                   | Method | Auth                   | Request Body / Params                           | Response                                                  |
|----------------------------|--------|------------------------|-------------------------------------------------|-----------------------------------------------------------|
| /api/v1/digitize           | POST   | JWT Bearer             | Multipart: PDF/PNG file upload                  | JSON: entities, CHS score, entity_count                   |
| /api/v1/claims             | POST   | JWT Bearer             | JSON: beneficiary_id, land_type, gps_polygon_id | JSON: claim_id, status: SUBMITTED                         |
| /api/v1/claims/{id}        | GET    | JWT Bearer             | Path: claim_id string                           | JSON: full claim record + DSS output                      |
| /api/v1/claims/{id}/status | PATCH  | JWT Bearer (SDLC role) | JSON: status, comments                          | JSON: updated claim record                                |
| /api/v1/gps                | POST   | JWT Bearer             | JSON: vertex_array (lat/long pairs), claim_id   | JSON: gps_polygon_id, area_acres, conflict_flag           |
| /api/v1/dss/evaluate       | POST   | JWT Bearer             | JSON: beneficiary_id, claim_id                  | JSON: vulnerability_score, convergence_score, scheme_list |
| /api/v1/schemes            | GET    | JWT Bearer             | Query: category (optional filter)               | JSON: array of 17 scheme profiles with eligibility rules  |
| /api/v1/nyayasetu/status   | GET    | Phone OTP              | Query: phone_number                             | JSON: claim_status, scheme_recommendations, patta_url     |

Table A3: API Endpoint Documentation

## Appendix D: Glossary of Terms

| Term / Abbreviation | Full Form / Definition                                                                          |
|---------------------|-------------------------------------------------------------------------------------------------|
| FRA                 | Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 |
| OCR                 | Optical Character Recognition — AI-based extraction of text from scanned images                 |
| NER                 | Named Entity Recognition — classification of named entities                                     |

| Term / Abbreviation | Full Form / Definition                                                                                                 |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
|                     | (person, location, quantity) in text                                                                                   |
| PostGIS             | Open-source spatial extension to PostgreSQL providing ST_Polygon, ST_Intersects, and GiST indexing                     |
| DSS                 | Decision Support System — Astitava-s eligibility and welfare scheme recommendation engine                              |
| NDVI                | Normalised Difference Vegetation Index — satellite measure of vegetation density from Sentinel-2 bands                 |
| SDLC                | Sub-Divisional Level Committee — government body reviewing FRA claims at sub-district level                            |
| DLC                 | District Level Committee — government body with final authority on FRA claim approval or rejection                     |
| Patta               | Land title document issued to a tribal claimant on successful FRA claim approval by DLC                                |
| PVTG                | Particularly Vulnerable Tribal Group — tribal communities with additional legal protections and priority scheme access |
| CHS                 | Claim Health Score — AI-generated completeness and consistency score (0.0–1.0) for each digitised FRA claim            |
| RBAC                | Role-Based Access Control — system enforcing data access based on authenticated user role and district scope           |
| ST                  | Scheduled Tribe — communities listed under Article 342 of the Constitution of India                                    |
| Van Adhikar Mitra   | Village-level field facilitator who assists tribal claimants in filing FRA claims using the Astitava Mobile App        |
| GPS                 | Global Positioning System — satellite-based geolocation system used for land boundary capture                          |
| DPDPA               | Digital Personal Data Protection Act, 2023 — India-s primary legislation governing personal data processing            |

Table A4: Glossary of Terms